

Formulation, Characterization, and Time Series Forecasting of a
Baseline Impact-Weighted Philippine Air Quality Index (BIWPhAQI)

by

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List of Abbreviations

Abbreviation	Definition
ARIMA	Autoregressive Integrated Moving Average
AQHI	Air Quality Health Index
AQI	Air Quality Index
CAMS	Copernicus Atmosphere Monitoring Service
CO	Carbon Monoxide
CCHAIN	Climate Change, Health, and Artificial Intelligence
ETS	Error, Trend, Seasonality
ERA5	European Centre for Medium-Range Weather Forecasts Reanalysis 5th Generation
HRAPIE-2	update of Health risks of air pollution in Europe (project)
MERRA-2	Modern-Era Retrospective analysis for Research and Applications, Version 2
NO ₂	Nitrogen Dioxide
NNAR	Neural Network Autoregressive
O ₃	Ozone
PSGC	Philippine Standard Geographic Code
PM _{2.5}	particulate matter, where particles have an aerodynamic diameter equal to or less than 2.5 µm
PM ₁₀	particulate matter, where particles have an aerodynamic diameter equal to or less than 10 µm
RR	Relative Risk
SO ₂	Sulfur Dioxide
WHO	World Health Organization

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Chapter I - Introduction

A. Background

Air pollution presents a major environmental and public health challenge worldwide. Increases in urbanization, economic activity, and energy use have significantly harmed air quality in both developed and developing countries (World Health Organization [WHO], 2021). Major air pollutants include fine and coarse particulate matter, nitrogen dioxide, sulfur dioxide, carbon monoxide, and ground-level ozone. These substances are strongly linked to harmful outcomes affecting the respiratory, cardiovascular, and developmental systems (Lelieveld et al., 2019). The pollutants interact through complex atmospheric pathways and their levels vary across time and location. This complexity makes risk assessment and policy implementation difficult.

The most concerning pollutants include fine particulate matter, which penetrate deep into the lungs, enter the bloodstream, and trigger oxidative stress, inflammation, and blood clotting responses. Long-term exposure to this consistently increases the risks of heart disease, stroke, lung cancer, and total mortality (Yang et al., 2022). In contrast, coarse particulate matter mainly causes upper and lower respiratory irritation and worsens conditions like asthma. Nitrogen dioxide is a marker of traffic emissions that causes airway inflammation and increases respiratory illness and death (Huang et al., 2021). Ground-level ozone is a secondary pollutant that injures airway cells, increasing bronchial sensitivity and contributing to respiratory hospitalizations (Donzelli et al., 2024). Sulfur dioxide causes breathing problems and contributes to the formation of secondary particles. Finally, carbon monoxide, an odorless gas from incomplete combustion, reduces oxygen delivery in the body (McMahon & Launico, 2025).

Since these pollutants do not act alone, health-focused monitoring should combine multiple pollutants to accurately account for their joint health impact (Cao et al., 2021).

Air Quality Indices (AQIs) are standardized tools that translate pollutant measurements into easy indicators of potential health risk. Major systems include the U.S. EPA's AQI, Canada's Air Quality Health Index (AQHI), and China's AQI. These systems differ in which pollutants they cover, how they weight them, and how they estimate health risk. The U.S. EPA AQI often uses the maximum sub-index, or "dominant pollutant", to set the daily value (Stieb et al., 2008). In contrast, health-based indices like Canada's AQHI combine the effects of multiple pollutants using coefficients from health data to estimate short-term population risk (Cao et al., 2021). AQIs differ in several ways, like depending on which pollutants are included, the statistical rule for combining them, the link to health outcomes, and whether they consider population exposure patterns. Health-based, multipollutant indices are better at capturing combined exposure risks, and they can be more useful for public health action. AQIs help summarize complex data, track air quality trends, and inform policy interventions. However, indices that fail to account for how populations are distributed or how pollutants interact may misrepresent the actual health impact.

B. Objectives

As a result of these gaps in the body of knowledge, the general objective of this study is to develop a time series model in pursuit of forecasting impact-weighted air quality index for the Philippines that can provide a foundation for future public health monitoring and risk assessment, in line with the principles of the sustainable development goals, specifically SDG3, SDG11, and SDG13. Specifically, this paper aims:

1. To construct a Philippine Air Quality Index using pollutant concentration data from the Project CCHAIN Climate and Air Quality Dataset, applying the transformed mean relative risks for long-term respiratory mortality as pollutant-specific weights.
2. To establish a Baseline Impact-Weighted Philippine Air Quality Index (BIWPhAQI) by generating a population-weighted composite time series based on the constructed air quality index and barangay-level population data.
3. To develop and compare multiple time series forecasting models for the BIWPhAQI, including ARIMA, ETS, and NNAR models, using training–test split validation to determine the model with the best goodness-of-fit and forecasting accuracy.
4. To produce a six-year forecast of the BIWPhAQI using the best-performing time series model.

C. Significance

This study supports the Sustainable Development Goals by creating an impact–weighted air quality index and forecasting system that can guide evidence-based environmental and public health actions. By linking air pollution to potential health risks, the research directly advances SDG 3 (Good Health and Well-Being), enhances urban environmental management under SDG 11 (Sustainable Cities and Communities), and contributes to climate-related planning aligned with SDG 13 (Climate Action).

D. Scope and Limitations

Several limitations should be acknowledged in interpreting the results of this study. First, the construction of the Baseline Impact-Weighted Philippine Air Quality Index relies on

non-local relative risk (RR) estimates for long-term respiratory mortality. The RR values used were drawn from international epidemiological studies such as HRAPIE-2 and other global meta-analyses rather than Philippine-specific cohort evidence. Because pollutant toxicity and health vulnerability vary by population, relying on foreign RR values may introduce bias into the weighting scheme and limit the precision of the index for Philippine conditions; nevertheless, this study establishes a solid groundwork that could help with future related studies. Second, the dataset is constrained by the availability of air quality and population data only until 2021, with subsequent values requiring extrapolation or interpolation due to missing population counts. This restricts the model's ability to capture post-2021 environmental, economic, and policy changes, which may have materially influenced air quality patterns. Third, the BIWPhAQI aggregates barangay-level data into a national index. Although population weighting helps address exposure variability, the aggregation may mask important spatial heterogeneities and local pollution episodes that behave differently from national averages. These limitations underscore the need for more localized epidemiological data, extended monitoring coverage, and improved ground-level validation. Lastly, because the index is constructed from weighted and covariance-adjusted pollutant concentrations, its scale does not map onto a widely recognized public health metric, making it challenging to interpret changes in practical terms; for example, what a 1-unit increase represents for real-world health risk or exposure conditions. These limitations underscore the need for more localized epidemiological data, extended monitoring coverage, more interpretable index calibration, and improved ground-level validation.

E. Outline

This study introduces the Baseline Impact-Weighted Philippine Air Quality Index, a novel, epidemiologically informed, population-weighted indicator designed to characterize air quality and forecast its changes over time. Leveraging barangay-level geocoded data, pollutant concentrations, and population estimates, the index integrates: (1) conversion of pollutant measurements to common units, (2) weighting of pollutants using relative risk-derived coefficients, (3) covariance-adjusted aggregation to reflect joint pollutant behavior, and (4) population weighting to reflect differential exposure across Philippine urban centers. To characterize the temporal properties of the resulting index, the study further employs decomposition techniques, outlier detection, and stationarity testing. Finally, this study conducts comparative time series forecasting using ARIMA, ETS, and NNAR models to evaluate the BIWPhAQI's predictability and trends.

Chapter II – Review of Related Literature

Forecasting plays a critical role in air quality management, enabling early warnings, resource allocation, and long-term planning. Traditional statistical models, particularly ARIMA and ETS, remain widely used because of their interpretability and strong performance in seasonal environmental data. Meanwhile, nonlinear and machine-learning approaches, such as Artificial Neural Networks (ANN), Extreme Learning Machines (ELM), and hybrid models, have become increasingly prominent for handling complex pollutant behavior.

The study by Hasan et al. (2024) in Dhaka provides recent and relevant evidence demonstrating the value of applying diverse forecasting techniques to particulate matter. Using real-world data from 2016 to 2022, the authors compared six models, namely ARIMA, ETS,

ANN, ELM, Naïve, and TBATS, and found that ARIMA performed best for $PM_{2.5}$ forecasting, while TBATS slightly outperformed others for PM_{10} . Their findings emphasize several important insights: (1) the persistence of traditional models such as ARIMA in structured environmental datasets; (2) the ability of machine-learning models to capture local nonlinearities; and (3) the need to evaluate multiple models using robust accuracy metrics such as RMSE, MAE, and MAPE. They also underscore the role of data preprocessing, proper training-testing splits, and the integration of meteorological factors in optimizing predictive performance. These conclusions align strongly with the objectives of the present study, which similarly compares ARIMA, ETS, and NNAR models to identify the most appropriate approach for forecasting the BIWPhAQI.

In addition, long-term exposure to fine particulate matter is widely recognized as a major driver of chronic respiratory and cardiovascular disease burden. A meta-analysis by Long-term exposure to particulate matter on cardiovascular and respiratory diseases in low- and middle-income countries by Guo et al. (2023) estimated that chronic exposure to $PM_{2.5}$ and PM_{10} respectively raises long-term respiratory mortality risk by about 30% and 28%, with older adults particularly vulnerable. In the Philippine context, it is estimated that ambient air pollution causes approximately 66,230 premature deaths annually, many of which stem from respiratory and pollution-related illnesses. Because fine particles like $PM_{2.5}$ penetrate deep into the lungs and can provoke long-term damage to respiratory function, long-term respiratory mortality relative risk can be used as an indicator of the health impact of a certain air pollutant.

With this, Cairncross, John, and Zunckel (2007) introduced a health-based air pollution index that moves beyond traditional breakpoint systems by directly incorporating epidemiologically derived relative risks. Instead of translating pollutant concentrations into

arbitrary categories, their index estimates the incremental daily mortality risk associated with short-term exposure to multiple pollutants. Using concentration–response relationships, each pollutant’s incremental risk is calculated based on its RR and deviation from baseline concentrations, and these risks are summed to produce a combined daily mortality risk value. This risk-based metric is then mapped to a simplified index scale for communication. By grounding the index in epidemiological evidence and accounting for combined pollutant exposure, the authors offer a more realistic, health-relevant alternative to conventional AQIs, which often overlook the cumulative nature of pollution and the presence of health effects even at concentrations below guideline thresholds.

The researchers’ framework has influenced subsequent efforts to develop impact-weighted air quality indicators that better reflect population exposure and public health risk. Their approach demonstrates how relative risks can be used as pollutant-specific weights to construct an index directly tied to health outcomes, providing a model for regions seeking more actionable air quality metrics. This is closely aligned with the present study, which similarly uses relative risks, though for long-term respiratory mortality, to construct the Baseline Impact-Weighted Philippine Air Quality Index. By integrating RR-based weighting with population adjustments and time series forecasting methods, the current work extends the core principles of their study toward a more localized and forward-looking tool for environmental health monitoring in the Philippines.

Chapter III – Data Series and Methods

A. Dataset

The primary dataset used in this research paper is the Philippine Climate and Air Quality Dataset, which was produced for Project CCHAIN (Climate Change, Health, and Artificial INtelligence) through the Lacuna Fund (Thinking Machines Data Science, Inc., 2024). Project CCHAIN was conceived to fill the gap in the public availability and accessibility of barangay-level data in the Philippines, as it integrates climate, environmental, socioeconomic, and health information from various government and institutional sources into one harmonized, analysis-ready resource. This undertaking was accomplished by a consortium that included Thinking Machines Data Science, EpiMetrics, Inc., the Manila Observatory, and the Philippine Action for Community-led Shelter Initiatives, which collectively harmonized multi-resolution datasets covering 879 barangays. Funded by the Lacuna Fund, a global effort to expand open, high-quality datasets for climate and health research in low- and middle-income contexts, this project aims to support researchers, policymakers, and local governments with the data required to analyze how climate and environmental factors determine health outcomes, especially in vulnerable and informal communities. By providing records at the barangay level, the dataset allows the making of fine-grained assessments and localized interventions that would otherwise be impossible to perform using broader, city-level records. Overall, the dataset is made up of 29 interrelated tables spanning 2003 to 2022 across 12 Philippine cities: Dagupan, Palayan, Navotas, Mandaluyong, Muntinlupa, Legazpi, Iloilo, Mandaue, Tacloban, Zamboanga, Cagayan de Oro, and Davao.

Data for the climate and air quality component of the dataset were derived from openly accessible global reanalysis products, which provide spatially consistent, daily observations

across the regions of interest. This dataset contains daily pollutant concentration estimates at the barangay level for nitrogen dioxide (NO₂), carbon monoxide (CO), sulfur dioxide (SO₂), ozone (O₃), particulate matter smaller than 10 µm (PM₁₀), and particulate matter smaller than 2.5 µm (PM_{2.5}). These indicators were sourced from three established atmospheric monitoring and reanalysis systems. NO₂, PM₁₀, and PM_{2.5} concentrations were obtained from the Copernicus Atmosphere Monitoring Service (CAMS), which combines satellite retrievals, ground-based measurements, and chemical transport modeling in order to produce physically consistent daily pollutant fields worldwide. Estimates for CO and SO₂ were supplied by NASA's Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2), a long-term gap-filled atmospheric dataset that assimilates satellite radiances, aerosol optical depth, and meteorological observations using the GEOS-5 Earth system model. Lastly, O₃ mixing ratios came from the ECMWF's 5th-Generation Reanalysis (ERA5), which assimilates an extensive array of ground, satellite, and radiosonde observations to produce hourly high-resolution global atmospheric fields.

In addition, population data were sourced from the Philippine WorldPop Population Dataset from Project CCHAIN to enable aggregation of barangay-level data to the national scale. WorldPop generates high-resolution, gridded population estimates by integrating census data, satellite imagery, and geospatial modeling to produce spatially detailed population distributions. In this study, annual population counts for the years 2000 to 2020 for each barangay were determined by summing all the population values of all WorldPop grid cells that fall within the corresponding barangay boundary and resulting in the variable `pop_count_total`. Table 3.1 lists a summary of all variables used in this study.

Table 3.1: Summary of Major Variables Used in the Analysis

Variable Name	Definition
adm4_pcode	Philippine Standard Geographic Code (PSGC) identifying each barangay.
date	Date of observation; recorded daily for air quality indicators and annually for population counts.
no2	Concentration of nitrogen dioxide (ppb).
co	Concentration of carbon monoxide (ppm).
so2	Concentration of sulfur dioxide (ppb).
o3	Concentration of ozone (ppb).
pm10	Concentration of particulate matter with a diameter less than 10 μm ($\mu\text{g}/\text{m}^3$).
pm25	Concentration of particulate matter with a diameter less than 2.5 μm ($\mu\text{g}/\text{m}^3$).
pop_count_total	Sum of all population count grids within the barangay.

B. Data Preprocessing

To construct the Baseline Impact-Weighted Philippine Air Quality Index, pollutant concentrations were integrated with barangay-level data into a unified framework that accounts for differences in population exposure and associated health impacts. Pollutant concentrations were standardized such that 1 $\mu\text{g}/\text{m}^3$ represents the effective unit, ensuring consistent weighting across all pollutants in subsequent analyses. This standardization was performed by converting concentrations based on the molecular weights of the respective gases. For methodological consistency, a constant temperature of 27°C and a pressure of 1 atm were assumed throughout the calculations.

Long-term respiratory mortality relative risks for each pollutant, presented in Table 3.2, were adopted as weighting factors. The mean RR values were subsequently standardized by converting them into β coefficients using the standard epidemiological transformation:

$$\beta_p = \frac{\ln(RR_p)}{\Delta C_p},$$

Where RR_p is the relative risk associated with pollutant p , ΔC_p is the exposure increment corresponding to RR , and β_p represents the per-unit contribution of pollutant p to the health-risk-weighted index. As such, the β coefficient represents the change in risk of long-term respiratory mortality for each 1 $\mu\text{g}/\text{m}^3$ increase in a pollutant.

Table 3.2: Mean Relative Risks for Long-Term Respiratory Mortality

Pollutant	Mean Relative Risk for Long-Term Respiratory Mortality	Data Source
no2	1.05 per 10 $\mu\text{g}/\text{m}^3$	HRAPIE-2
co	1.0318 per 1 mg/m^3	Guo, et al. (2022)
so2	1.0067 per 10 $\mu\text{g}/\text{m}^3$	Orellano, et al. (2016)
o3	1.5 per 10 $\mu\text{g}/\text{m}^3$	HRAPIE-2
pm10	1.12 per 10 $\mu\text{g}/\text{m}^3$	HRAPIE-2
pm25	1.14 per 10 $\mu\text{g}/\text{m}^3$	HRAPIE-2

After standardizing pollutant concentrations and health-risk weights, it can be observed that pollutant levels are correlated. This is expected because many pollutants share common sources or undergo atmospheric chemical transformations that produce multiple pollutants simultaneously (Guo et al., 2020). Conventional weighting procedures that sum or average pollutant contributions risk overestimating overall health risk if these overlaps are ignored, effectively double-counting shared effects. As such, to correct for this, adjustment of the standardized β coefficients is done using the covariance structure of the pollutants as follows:

$$C = W\Sigma W, AdjIndex_i = \sqrt{(X_i C) \cdot X_i^T}$$

where X_i is the vector of pollutant concentrations at observation i , W is the diagonal matrix of standardized β coefficients, and Σ is the covariance matrix of the pollutants. By doing so, the index reflects a more independent contribution of each pollutant to health risk. This adjustment makes sure the index does not exaggerate health risks when multiple pollutants come from the same source or happen at the same time. However, doing so makes direct interpretation of the index blurry, and hence it can only be concluded that each 1-unit increase in the adjusted index reflects an increase in overall health risk from air pollution, accounting for both the strength and correlation of each pollutant's effect.

For the spatial component, population counts were used as aggregation weights to capture differential exposure across barangays. Because daily population data were unavailable, annual estimates from the WorldPop dataset were used, with linear interpolation applied to fill missing values for 2021 and 2022. For each day, the barangay-level adjusted indices were combined into a single national value by computing a population-weighted average, producing the daily Philippine impact-based AQI. These daily values were then averaged by month to generate the monthly time series.

C. Preliminary Diagnostics

The overall behavior of the BIWPhAQI was examined prior to modelling. A plot of the time series was visually inspected to see the variation of the index with time. Furthermore, the mean, skewness, and kurtosis were calculated to understand the typical level, the degree of asymmetry, and whether it is heavy-tailed. Outliers were also to be identified due to possible major influence of the results, and since these extreme values may be indicative of unusual

pollution events or measurement issues. The *tsoutliers* function was used to identify these anomalous observations, which employs *supsmu* for non-seasonal components and a periodic STL decomposition for seasonal structure to detect outliers and generate appropriate replacement estimates.

Additive, multiplicative, and STL methods were then used to decompose the series into three components, namely the trend, seasonality, and the error, to determine any noticeable patterns. A seasonal plot was also generated to highlight systematic differences across months or seasons. The autocorrelation and partial autocorrelation functions (ACF and PACF) in order to see if the level of pollution on a certain day is influenced by repeating lag structures. Stationarity was also tested through the KPSS unit root test. Transformations like differencing and the Box-Cox method will be used to stabilize the series and prepare it for the use of time series forecasting models.

D. Time Series Modelling

The dataset will be split into a training set and a test set. The training set will consist of AQI data from January 2003 to December 2020, while the test set will include data from January 2021 to December 2022. This particular split point was chosen to clearly separate the period before and during the most drastic changes brought upon by the COVID-19 pandemic. By training the model on pre-pandemic data and testing on post-lockdown data, the models can learn from the typical air quality conditions and be checked to see if they adapt to the significantly altered environmental conditions. After fitting each model, residual diagnostics will be performed. In particular, the Ljung-Box test will be used to assess the presence of autocorrelation in the residuals, ensuring model adequacy.

Three classes of models will be applied to explore the data: SARIMA (Seasonal AutoRegressive Integrated Moving Average), ETS (Error, Trend, Seasonality), and NNAR (Neural Network AutoRegressive).

1. **SARIMA:** In identifying the best S/ARIMA model, candidate models were first classified. For the selection of the candidate models, both automatic and manual procedures were employed. For the automatic procedures, two methods were employed: using the `auto.arima()` function in R, which utilizes the Hyndman-Khandakar algorithm, and implementing the exhaustive algorithm, which is in R also. The models that were provided by these two methods were automatically included in the roster of candidate models, and the manual procedure was done next.

For the manual procedure, the ACF and PACF were used to select reasonable values for p and q , as well as their seasonal counterparts P and Q . For the parameters d and D , they were selected based on the plot of the time series, as well as the results of the KPSS test. All in all, three candidate models were selected using the manual procedure, resulting in a total of five candidate models. These five S/ARIMA models were then fitted, and the corresponding AICs were obtained and compared to each other; the model yielding the lowest AIC was then deemed the best SARIMA model.

2. **ETS:** In selecting the best ETS model, only an automatic procedure was employed through the built-in `ets()` function in R. The `ets()` function fits exponential smoothing state space models by automatically selecting the best combination of error (additive or

multiplicative), trend (none, additive, or damped), and seasonal components (none, additive, or multiplicative) using the AICc criterion. The ETS model provided by this function was automatically deemed the best ETS model, and was chosen to be compared to the others (SARIMA and NNAR).

3. **NNAR:** To select the best NNAR model, two procedures were conducted, which are also the manual and automatic procedures. The manual procedure was done first, which is done by fitting SARIMA(p,0,0)(1,0,0) models to the data, where $p = 1, \dots, 7$. $P = 1$ was chosen because it is the default value when considering a seasonal component when finding the optimal parameters for NNAR(p, P, k). Among the seven SARIMA models, the parameter p of the model with the lowest AIC was then applied to the NNAR model. Then, k was determined by computing $(p + P + 1) / 2$ and rounding up. The resulting NNAR(p, P, k) model was then deemed as a candidate model.

On the other hand, the automatic procedure was done by simply using the `nnetar()` function in R. The model provided by the automatic procedure was then found out to be exactly the same as the one identified by the manual procedure; and thus this model was immediately deemed the best NNAR model.

E. Forecasting

The fitted ARIMA, ETS, and NNAR models will be tested by comparing two-year forecasts (2021-2022) and then generating forecasts for an additional four years (2023-2026). Forecasting until 2026 provides information for possible long-term strategic planning. Forecast

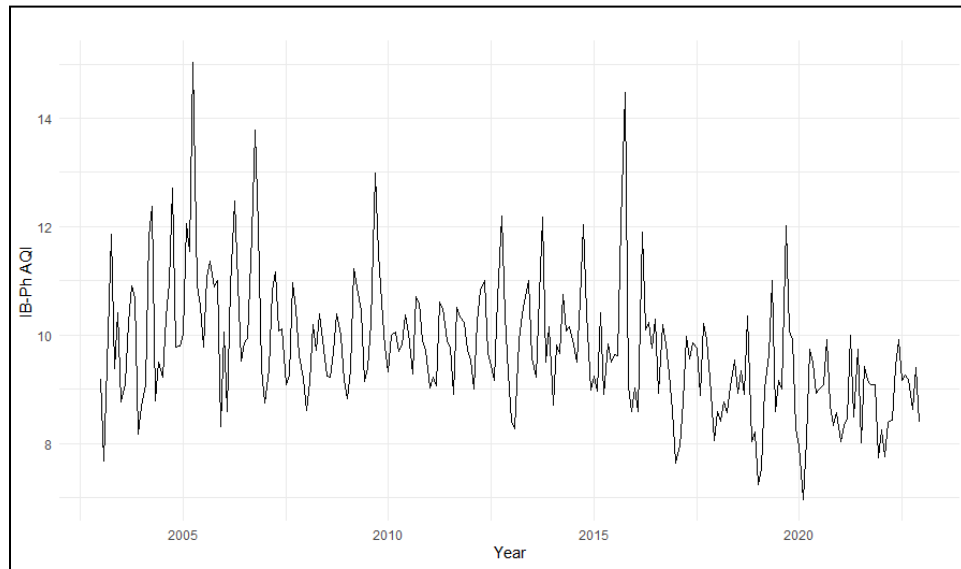
accuracy will be evaluated by comparing the predicted values to the test set observations. Standard error metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Scaled Error (MASE), will be calculated to compare model performance both in the training and test periods. These metrics will inform which model best captures the underlying patterns of the AQI series for reliable forecasting.

Chapter IV – Analysis of Results

A. Baseline Impact-Weighted Philippine Air Quality Index

After preprocessing the series and combining pollutant measurements into a single unified index, the BIWPhAQI time-series plot is presented in Figure 4.1. Overall, no clear long-term directional trend is manifested in the index, though a gradual downtrend can be seen toward the last years of the series. The plot is characterized by periodic oscillations, with noticeable peaks and troughs indicative of seasonal fluctuations in pollutant impact. While several sudden shifts were evident, likely influenced by episodic environmental events, the pattern appears rather constant and suggests that at least large-scale variations in air quality are governed by recurring atmospheric and seasonal cycles.

Figure 4.1: Baseline Impact-Weighted Philippine Air Quality Index

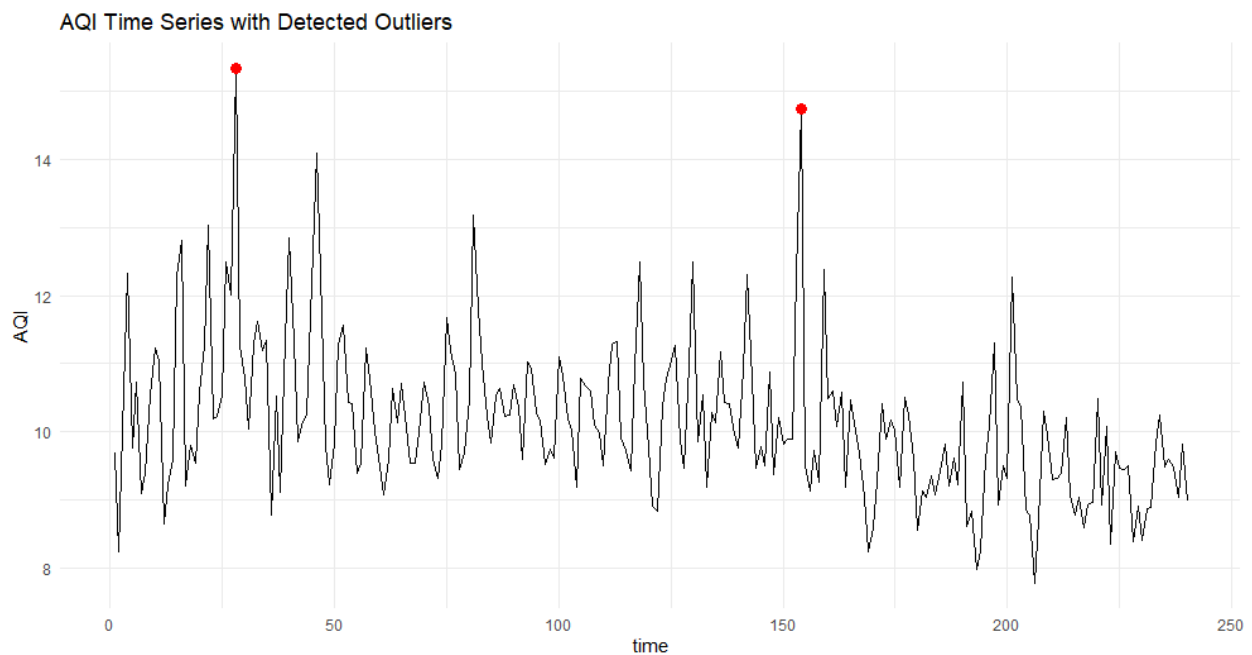


The distributional characteristics of the BIWPhAQI summarized in Table 4.2, show values ranging from approximately 7.8 to 15.3, with a mean slightly above 10. The distribution is right-skewed and leptokurtic, reflecting that most of the observations bunch around the lower-to-middle values of the series, with higher-than-usual spikes every now and then. Variability remains modest, as reflected by the standard deviation of about 1.13. The minimum index value for February 2020 coincides with the early COVID-19 lockdown period when mobility, transportation activity, and industrial operations fell substantially during a period of time known to temporarily improve air quality. Conversely, the highest value recorded in April 2005 coincides with the Anatahan volcanic eruption, which released large amounts of particulate matter into the atmosphere. Two outliers can be seen in Figure 4.3, for April 2005 and October 2015. These observations were kept in the analysis as they reflect actual environmental events, such as the 2015 Southeast Asian haze, and their presence allows for a more realistic representation of extreme but relevant atmospheric conditions.

Table 4.2 Distribution of BIWPhAQI

Min	1st Qu.	Median	Mean	3rd Qu.	Max
7.790	9.423	10.010	10.157	10.718	15.320
02 / 2020	08 / 2012	07 / 2017	N/A	3 / 2012	04 / 2005
Skewness		Kurtosis		Standard Deviation	
1.156		5.540		1.135	

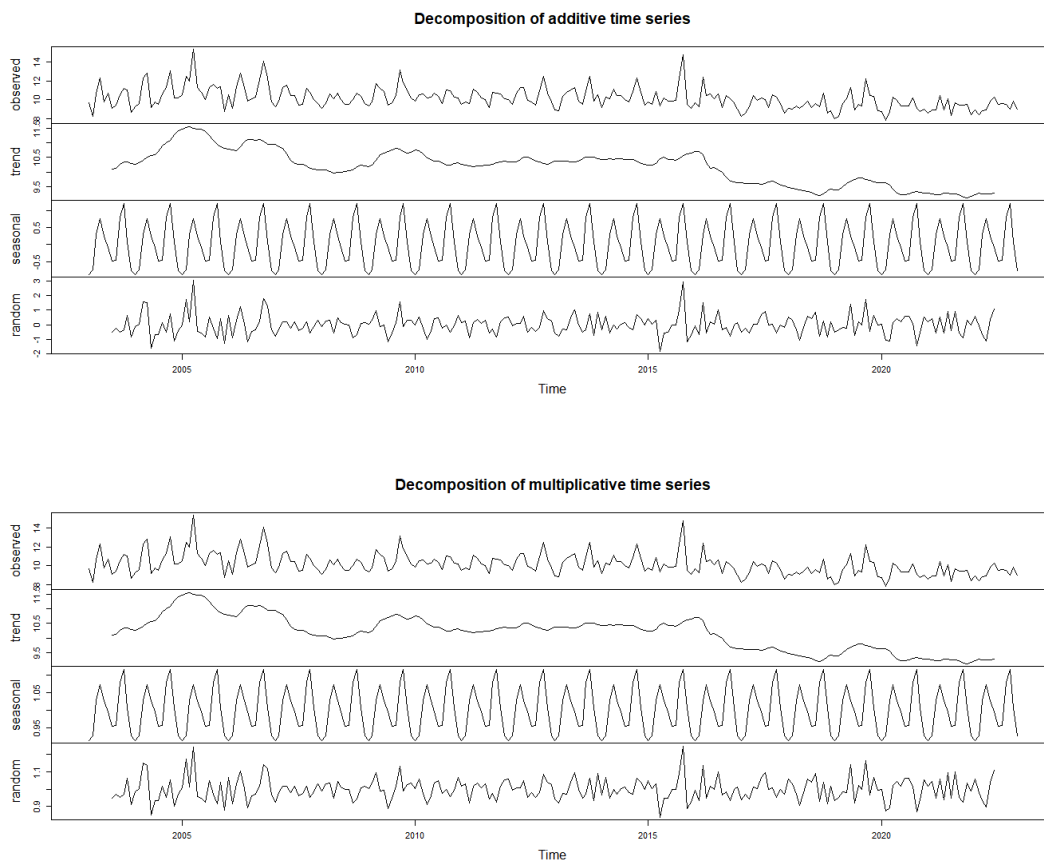
Figure 4.3. Outlier points in the BIWPhAQI plot

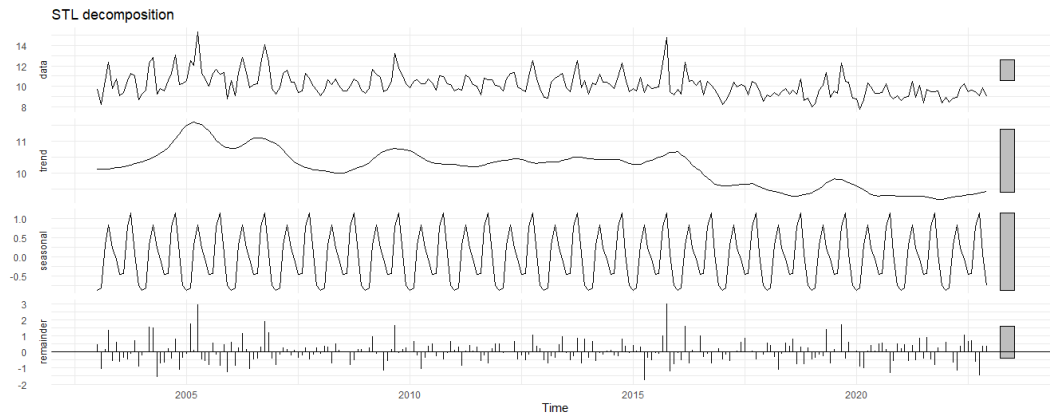


The time-series decomposition using additive, multiplicative, and STL methods represented in Figure 4.4 reveals a generally consistent structure across the three techniques. The trend component exhibits a gradual upward shift until about 2005, after which it slowly declines over the next few years. The seasonal patterns are quite strong and definitely indicate the

presence of predictably cyclic behavior of the index. The remainder components are small and rather consistent, indicating that all three decomposition approaches capture the essence of this series without undue variability.

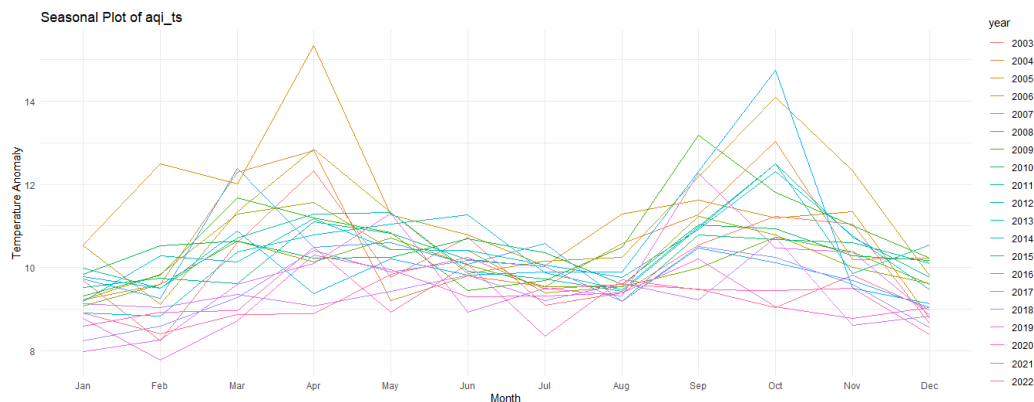
Figure 4.4: Decomposition of BIWPhAQI





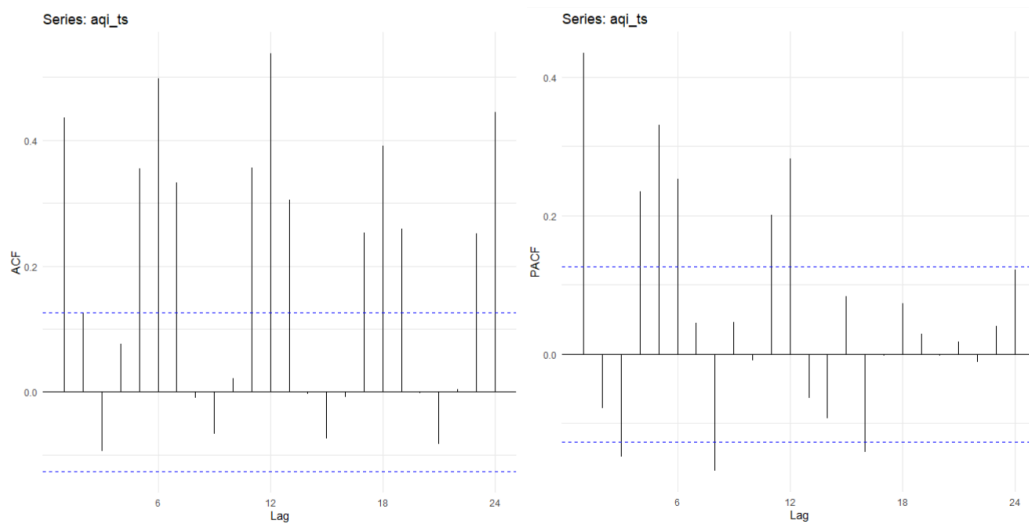
The seasonal plot shown in Figure 4.5 further portrays the concentration and dispersion of values across the annual cycle. Variability increases around April and October, months corresponding to transitional periods between the dry and wet seasons in the Philippines, when shifting meteorological conditions can heighten pollutant accumulation or dispersion. In contrast, the seasonal bands become tighter around December and June, indicating more stable atmospheric behavior in these months.

Figure 4.5: Seasonal Plot of BIWPhAQI



The autocorrelation and partial autocorrelation patterns shown in Figure 4.6 reflect strong correlations at low lags, with significant seasonal spikes, indicating that the BIWPhAQI reflects both recent observations as well as recurring seasonal patterns. In addition, for purposes of time-series modeling, stationarity of the series was assessed. Using a Box-Cox transformation (λ close to -1 , indicating a reciprocal transformation) followed by a first order differencing produced a series with a constant mean and variance. The KPSS test output in Figure 4.7 reveals that the transformed and differenced series is essentially stationary, and hence prepared for additional model development.

Figure 4.6: Autocorrelation and Partial Autocorrelation Functions of BIWPhAQI



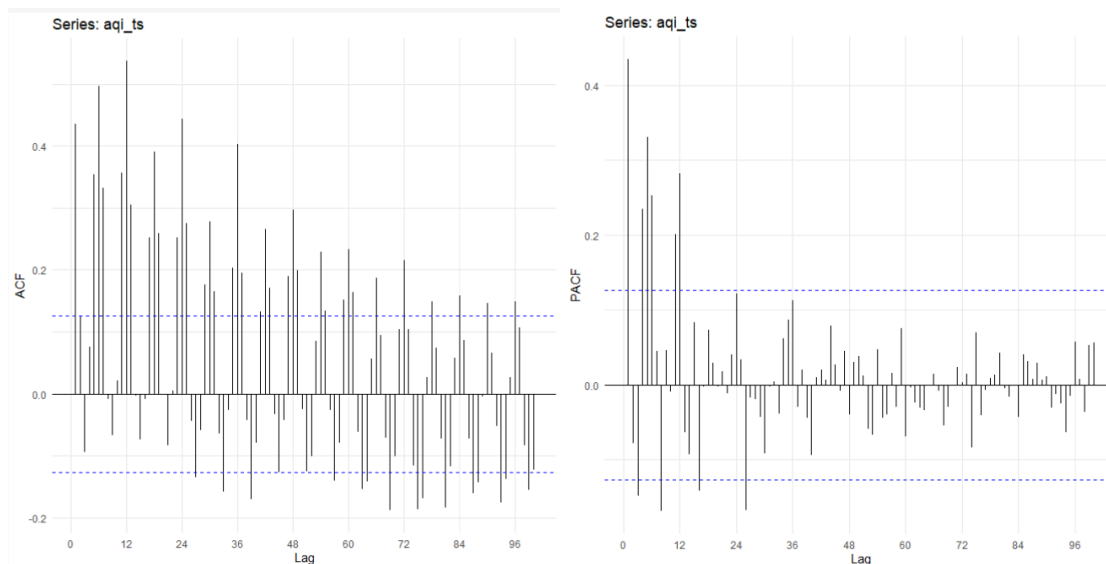


Table 4.7: KPSS Unit Root Test of Differenced and BoxCox Transformed BIWPhAQI

```
> aqi_transformed <- BoxCox(aqi_ts, lambda = BoxCox.lambda(aqi_ts))
> summary(ur.kpss(diff(aqi_transformed)))
```

```
#####
# KPSS Unit Root Test #
#####
```

```
Test is of type: mu with 4 lags.
```

```
value of test-statistic is: 0.0229
```

```
critical value for a significance level of:
```

```
      10pct  5pct 2.5pct  1pct
critical values 0.347 0.463 0.574 0.739
```

B. Time Series Modelling

SARIMA

The modelling procedure was started with the automatic selections of SARIMA models, and the results are as follows:

Table 4.8: auto.arima() results in R; SARIMA(1, 0, 2)(2, 1, 0)[12]

```
Series: train
ARIMA(1,0,2)(2,1,0)[12] with drift
Box Cox transformation: lambda= -0.9999242

Coefficients:
      ar1      ma1      ma2      sar1      sar2      drift
    -0.4041  0.6941  0.3769  -0.6429  -0.4121  -1e-04
s.e.    0.2561  0.2463  0.0713   0.0708   0.0740   1e-04

sigma^2 = 5.307e-05: log likelihood = 713.98
AIC=-1413.96  AICc=-1413.38  BIC=-1390.73
```

Table 4.9: Algorithm code results; SARIMA(2, 0, 0)(1, 0, 1)[12]

```
Series: train
ARIMA(2,0,0)(1,0,1)[12] with non-zero mean
Box Cox transformation: lambda= -0.9999242

Coefficients:
      ar1      ar2      sar1      sma1      mean
    0.2519  0.2178  0.9608  -0.6944  0.9000
s.e.    0.0680  0.0694  0.0220   0.0744  0.0039

sigma^2 = 5.347e-05: log likelihood = 752.13
AIC=-1492.25  AICc=-1491.85  BIC=-1472
```

The SARIMA model given by auto.arima() is SARIMA(1,0,2)(2,1,0)[12], while the model suggested by the algorithmic code is SARIMA(2,0,0)(1,0,1)[12]. These two were undoubtedly deemed as candidate models, and the manual procedure was immediately started. Recalling that the KPSS yielded a significant result, this signifies that the series is not stationary, and thus d must be set to be equal to 1 to conduct first-order differencing. Then, investigating the PACF (from figure 4.5), a spike can be seen at lag 1, while lag 2 stayed within the bounds, thus it can be argued that it cuts off at the first lag. Given the behavior of the ACF (from figure 4.5) which is decaying in nature, it is deemed appropriate for p to be equal to 1. Further, looking at the ACF, we see that there is also a large spike in lag 1 then it cuts off, and given the

decreasing nature also of the PACF, this constitutes that q must be 1. For the seasonal components, it can clearly be seen in the PACF that there is a clear spike at lag 12, and none at lags 24, 36, 48, and so on. Meanwhile, looking at the ACF in lags 12, 24, 36, and multiples of $m = 12$, it can be observed that it is showcasing decay. This is a clear sign that the parameter $P = 1$. D was then set to 0 because the researchers did not see the need for seasonal differencing since there are no clear signs of an unstable seasonal pattern, and although Q was set to 0 at first, it will be explored later on. Thus, the third candidate model is SARIMA(1, 1, 1)(1, 0, 0)[12]; this model was fitted and the results are as follows:

Table 4.10: Fitting SARIMA(1, 1, 1)(1, 0, 0)[12]

```
Series: train
ARIMA(1,1,1)(1,0,0)[12]
Box Cox transformation: lambda= -0.9999242

Coefficients:
      ar1      ma1      sar1
      0.2511  -0.9635  0.4707
s.e.    0.0724   0.0234  0.0643

sigma^2 = 6.652e-05:  log likelihood = 728.16
AIC=-1448.33  AICC=-1448.14  BIC=-1434.84
```

For the fourth candidate SARIMA model, the change was minimal. Looking at the PACF again, it can be noticed that after the spike at the first lag, it is true that the second lag stayed within the bounds. However, it can also be seen that there are consistent spikes at lags 3, 4, 5, and 6, then it cuts off. Hence, for the fourth model, the parameter p was let to be 6, resulting in a model of SARIMA(6, 1, 1)(1, 0, 0)[12], whose results are below:

Table 4.11: Fitting SARIMA (6, 1, 1)(1, 0, 0)[12]

```

Series: train
ARIMA(6,1,1)(1,0,0)[12]
Box Cox transformation: lambda= -0.9999242

Coefficients:
          ar1      ar2      ar3      ar4      ar5      ar6      ma1      sar1
      0.2689  0.1687 -0.3052  0.0969  0.2143  0.0642 -0.9771  0.4211
s.e.  0.0710  0.0771  0.0721  0.0747  0.0715  0.0753  0.0194  0.0810

sigma^2 = 5.851e-05: log likelihood = 744.56
AIC=-1471.12  AICC=-1470.24  BIC=-1440.78

```

For the fifth and last SARIMA model, a parameter in the seasonal component was modified. Since there are consistent spikes at lags 12, 24, 36, and so on, of the ACF, 1 order of seasonal MA was included, and all the other parameters were retained. Hence, the last candidate model is SARIMA(6, 1, 1)(1, 0, 1)[12].

Table 4.12: Fitting SARIMA(6, 1, 1)(1, 0, 1)[12]

```

Series: train
ARIMA(6,1,1)(1,0,1)[12]
Box Cox transformation: lambda= -0.9999242

Coefficients:
          ar1      ar2      ar3      ar4      ar5      ar6      ma1      sar1      sma1
      0.2371  0.1960 -0.1972  0.1232  0.1902  0.0467 -0.9870  0.9517 -0.7100
s.e.  0.0711  0.0742  0.0755  0.0745  0.0728  0.0730  0.0168  0.0304  0.0838

sigma^2 = 5.009e-05: log likelihood = 758.53
AIC=-1497.06  AICC=-1495.99  BIC=-1463.36

```

After getting all the candidate models, the next thing to do is to determine which of the five is the best by using an objective metric, which was chosen to be AIC. Upon doing so, the results are:

Table 4.13: AICs of the Five Candidate SARIMA Models

<code>> sapply(models, AIC)</code>					
	m1	m2	m3	m4	m5
	-1413.956	-1492.252	-1448.327	-1471.121	-1497.065

Where,

m1 = SARIMA(1, 0, 2)(2, 1, 0) [auto.arima()]

m2 = SARIMA(2, 0, 0)(1, 0, 1) [algorithmic code]

m3 = SARIMA(1, 1, 1)(1, 0, 0)

m4 = SARIMA(6, 1, 1)(1, 0, 0)

m5 = SARIMA(6, 1, 1)(1, 0, 1)

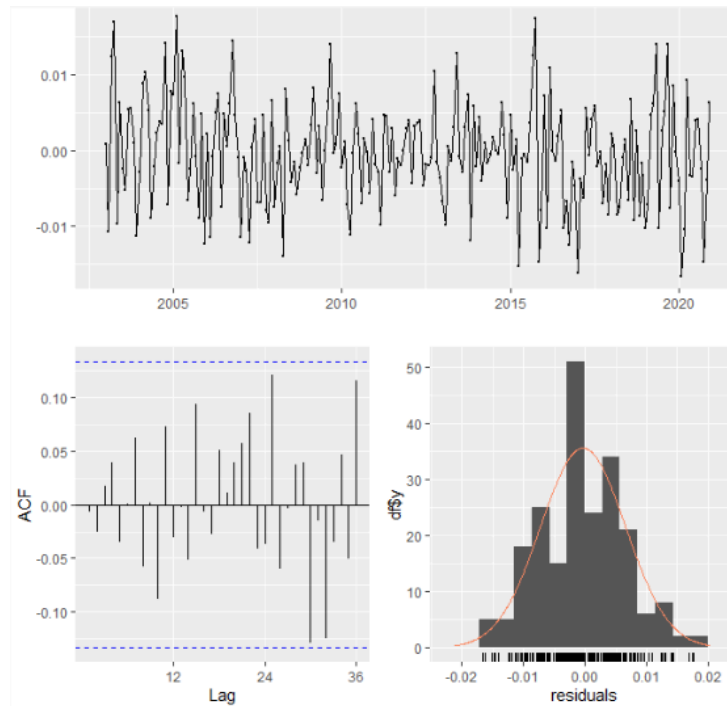
By basing on their AICs, it can be concluded that SARIMA(6, 1, 1)(1, 0, 1)[12] is the best since it has the lowest AIC among the five candidate models. Hence, the best SARIMA model includes up to 6 lags, order 1 differencing, 1 lagged error, 1 seasonal lag, and 1 seasonal lagged error. The written model is:

$$\begin{aligned}
 & (1 - 0.24B - 0.20B^2 + 0.20B^3 - 0.12B^4 - 0.19B^5 - 0.05B^6)(1 - 0.95B^{12})Y_t \\
 & = (1 - 0.99B)(1 - 0.71B^{12})e_t
 \end{aligned}$$

where Y_t = Box-Cox transformed X_t

After that, a residual analysis was conducted, and the results are:

Figure 4.14: Residual Analysis on SARIMA(6, 1, 1)(1, 0, 1)[12]



Ljung-Box test

```
data: Residuals from ARIMA(6,1,1)(1,0,1)[12]
Q* = 12.896, df = 15, p-value = 0.6103
```

```
Model df: 9. Total lags used: 24
```

The residual plots do not reveal any remaining structure in the fitted SARIMA model. The time-series plot shows residuals fluctuating randomly around zero, while the ACF displays only small, insignificant spikes, indicating no systematic autocorrelation left in the data. The histogram appears roughly symmetric and bell-shaped, suggesting that the residual distribution is reasonably well-behaved. The Ljung–Box test supports these visual observations, with a p-value of 0.6103 indicating no evidence of autocorrelation in the residuals. Overall, these diagnostics show that the model is adequate.

ETS

Moving on to the ETS model, the ets() function was utilized, and the results are follows:

Table 4.15: ETS Model Given by ets() Function in R

```
ETS(A,N,A)

Call:
ets(y = train, lambda = BoxCox.lambda(train), biasadj = T)

Box-Cox transformation: lambda= -0.9999

Smoothing parameters:
  alpha = 0.1456
  gamma = 1e-04

Initial states:
  l = 0.9027
  s = -0.007 8e-04 0.0108 0.0077 -0.0044 -0.0036
      -0.001 0.003 0.0075 0.0038 -0.0085 -0.0092

sigma: 0.0071

      AIC      AICC      BIC
-963.3632 -960.9632 -912.7340
```

Based on the results, the optimal ETS model is ETS(A, N, A). This means that the model has additive errors, no trend component, and an additive seasonal component. Further, the computed parameters are 0.1465 for alpha and 0.0001 for gamma. The written model is:

$$y_t = l_{t-1} + S_{t-m} + e_t,$$

$$l_t = l_{t-1} + 0.1456e_t,$$

$$s_t = s_{t-m} + 0.0001e_t,$$

$$\varepsilon_t = y_t - (l_{t-1} + S_{t-m})$$

Where,

y_t = *observed value*

l_t = *level*

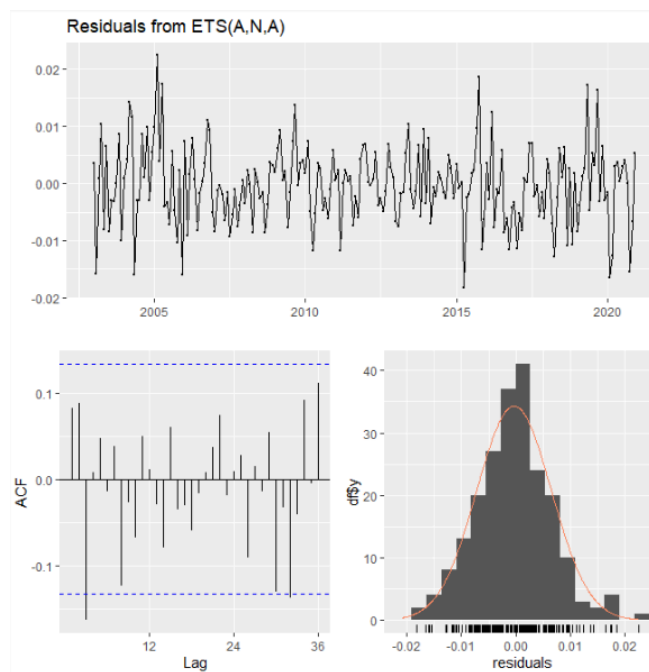
$s_t = \text{seasonal component}$

$m = \text{seasonal period}$

$\varepsilon_t = \text{additive error term}$

Since gamma is small, s_t is approximately equal to s_{t-m} , indicating a constant seasonality component. Further, the initial states are also listed above. The residual analysis was then performed, and the results are:

Figure 4.16: Residual Analysis on ETS(A, N, A)



Ljung-Box test

data: Residuals from ETS(A,N,A)
 $Q^* = 20.99$, $df = 24$, $p\text{-value} = 0.6393$

The residual graphs do not show any concerning patterns: the time plot looks random without noticeable trend or seasonality, the ACF bars are small and mostly within the bounds, and the histogram appears reasonably symmetric with a bell-shaped form. The Ljung–Box test reinforces this, with a p-value of 0.6393 indicating no evidence of remaining autocorrelation in the residuals. Taken together, these results suggest that the model performs well and can be considered adequate.

NNAR

Now tackling NNAR, starting with the manual selection, several candidate models were identified using SARIMA(p, 0, 0)(1, 0, 0)[12], where $p = 1, \dots, 7$, noting that $P = 1$ is default for seasonality; these SARIMA models were tested against each other using AIC. The results of this comparison are:

Table 4.17: AICs of SARIMA(p, 0, 0)(1, 0, 0)[12] for NNAR Manual Selection

AR1	AR2	AR3	AR4	AR5	AR6	AR7
-1454.623	-1456.941	-1464.492	-1471.487	-1481.396	-1480.286	-1478.726

The results showed that SARIMA(5, 0, 0)(1, 0, 0)[12] has the lowest AIC among them. Extracting p and P from this SARIMA model, the corresponding NNAR model is NNAR(5, 1, 4), where k is computed as $(5 + 1 + 1)/2$, then rounded up. Thus, this is the first NNAR candidate model.

Moving on to the automatic selection, the `nnetar()` function in R suggested the exact same NNAR model, as shown here:

Figure 4.18: Automatic nnetar() Results; NNAR(5, 1, 4)

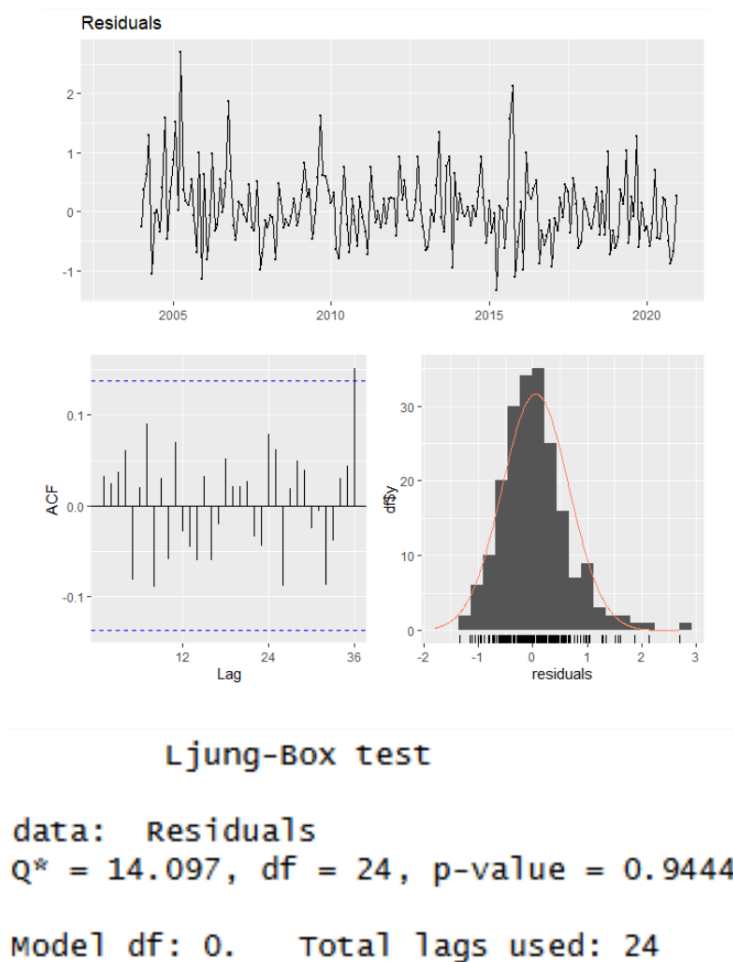
```
Model: NNAR(5,1,4)[12]
Call: nnetar(y = train, lambda = BoxCox.lambda(train), biasadj = T)

Average of 20 networks, each of which is
a 6-4-1 network with 33 weights
options were - linear output units

sigma^2 estimated as 3.831e-05
```

With results colliding, NNAR(5, 1, 4) was automatically deemed best by default. This model uses up to 5 lags, 1 seasonal lag, and 4 nodes in the hidden layer. Moreover, the residual analysis results are as follows:

Figure 4.19: Residual Analysis on NNAR(5, 1, 4)



The residual graphs show no remaining structure: the time plot exhibits randomness with no visible trend or seasonality, the ACF shows only small autocorrelations within the significance bounds, and the histogram is reasonably bell-shaped, indicating that the residuals behave approximately normally. The Ljung–Box test further supports this, with a very high p-value (0.9444), meaning it fails to reject the null hypothesis of no autocorrelation and conclude that the residuals resemble white noise. Taken together, the graphical diagnostics and statistical test indicate that the model is adequate.

C. Forecasting

Figure 4.18 displays the long-term training data alongside the 24-month forecast horizon. The data exhibits clear seasonality with recurring peaks and troughs, characteristic of air quality fluctuations driven by seasonal meteorological changes. Figure 4.19 provides a closer examination of the validation period from 2021-2022. The "test" series represents the actual observed index while the three other lines are the forecasts of the different models. The ARIMA and ETS forecasts closely mirror each other, suggesting that the linear components of the series, like trend and seasonality, are interpreted similarly by both statistical methods. However, they appear to struggle with the amplitude of the fluctuations in the test period. The NNAR forecast tracks the actual observations more dynamically. While it slightly under-predicts the highest peak in early 2021, it captures the subsequent downward trend and the oscillation patterns in late 2021 and 2022 more accurately than the linear models.

Figure 4.20: Forecasting of the Three Models from 2003 to 2022

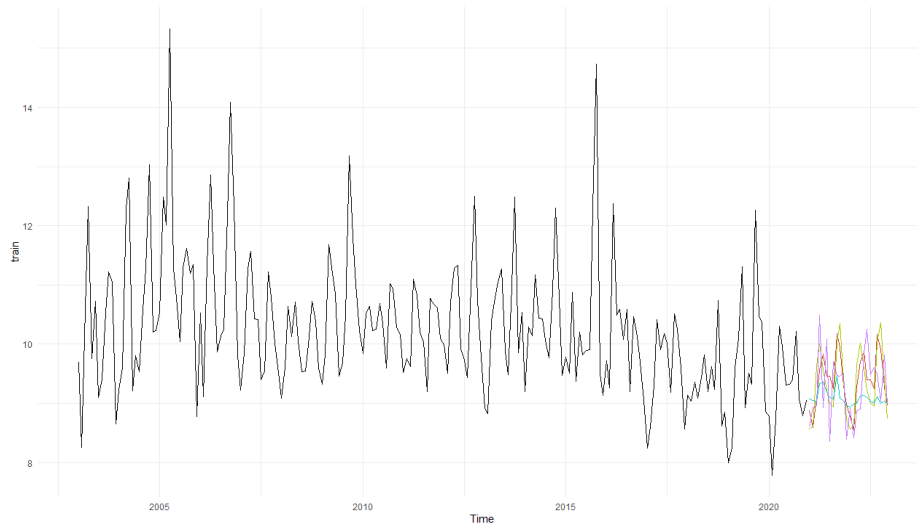
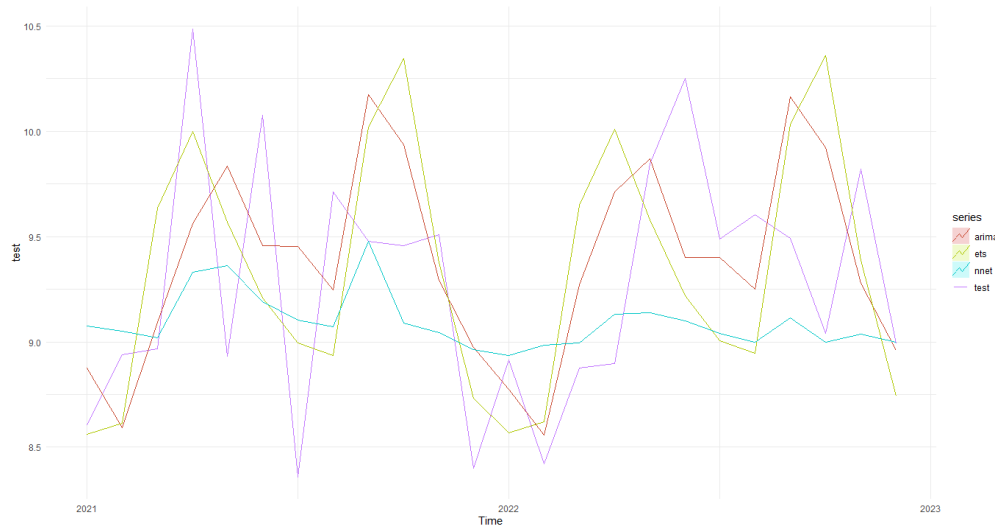


Figure 4.21: Forecasting of the Three Models from 2021 to 2022



This is supported by what is seen in Table 4.20, that the Neural Network model demonstrated superior predictive capability compared to the traditional statistical baselines. The NNAR model yielded an RMSE of 0.567, significantly lower than both the ETS (0.654) and ARIMA (0.578) models, indicating a tighter fit to the observed variances in the test period. In

terms of percentage error, the NNAR model maintained a MAPE of 4.82%, whereas the ARIMA and ETS models resulted in higher error rates of 5.24% and 6.17%, respectively. The Mean Absolute Scaled Error for the NNAR was 0.570, well below the threshold of 1.0, suggesting the model provides significant improvement over a naive random walk forecast. In contrast, the ARIMA and ETS models yielded MASE values of 0.608 and 0.718. It is worth noting that all three models performed better on the test set than the training set. This implies that the volatility or structural patterns in the 2021-2022 period were well-captured by the underlying trends identified during the training phase.

Table 4.22: Forecasting Accuracy of the Three Models

```
> accuracy(fc1,test);accuracy(fc2,test);accuracy(fc3,test)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.03584356	0.7896980	0.5970711	-0.8744744	5.733796	0.7468520	0.009961235	NA
Test set	-0.10387626	0.5780257	0.4859299	-1.3789713	5.239313	0.6078299	-0.391323861	0.6950575
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.03969209	0.7835906	0.5754535	-0.8963785	5.510217	0.7198114	0.09117499	NA
Test set	-0.06472514	0.6542441	0.5738952	-0.9426118	6.165383	0.7178622	-0.11271135	0.8080893
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.05773415	0.5922397	0.4417204	0.1240325	4.198642	0.5525301	0.02749281	NA
Test set	0.17948361	0.5665867	0.4560716	1.5954102	4.822730	0.5704814	-0.17050798	0.6878627

Overall, the comparative analysis suggests that the underlying process of the population-weighted air quality index contains non-linear dynamics that are best captured by the Neural Network Autoregression model. While ARIMA and ETS provided reasonable benchmarks, the NNAR is the most robust choice for forecasting future air quality levels in this context.

As illustrated in the Figure 4.22, the NNAR model exhibits a phenomenon known as rapid mean reversion. Approximately 12 to 18 months into the forecast period, the model fails to propagate the seasonal signal, with the predicted values collapsing into a flat trajectory near the long-run average. In the context of environmental data, specifically the Air Quality Index, this

behavior is theoretically implausible. AQI is driven by persistent meteorological cycles and patterns that do not simply vanish over time. Consequently, the NNAR model, while accurate in the short term, lacks the structural constraints necessary to simulate the cyclical volatility required for long-term planning.

The ARIMA model demonstrates superior utility for long-range forecasting. It successfully encodes the stochastic seasonality of the series. The plot confirms this as the ARIMA forecast maintains the oscillation between seasonal peaks and troughs through 2026. This is mechanically ensured by the seasonal component of the model, which explicitly forces the periodic structure into future estimates.

Figure 4.23: Forecasting of the Three Models from 2003 to 2026

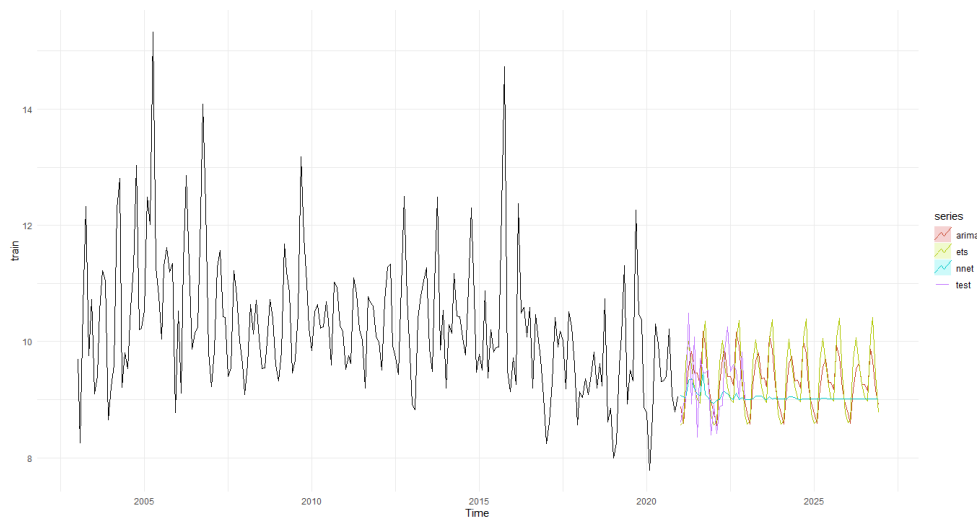
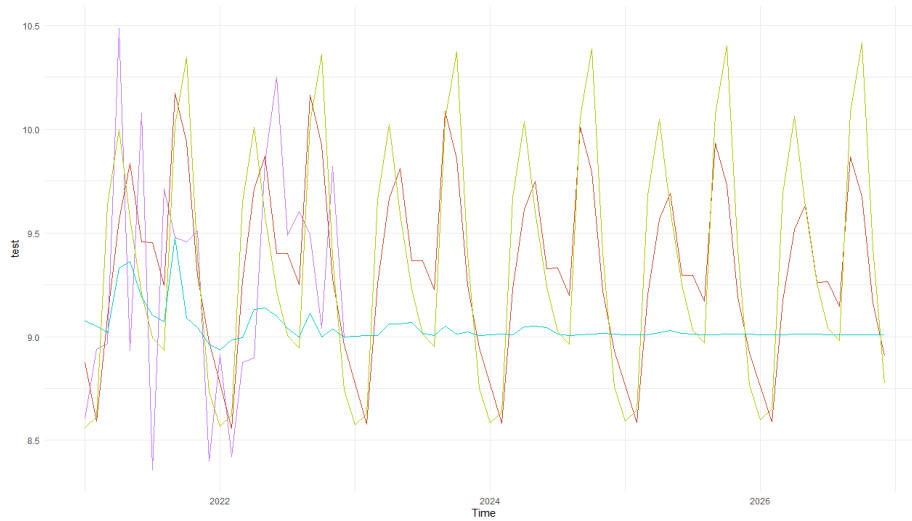


Figure 4.24: Forecasting of the Three Models from 2021 to 2026



In environmental monitoring, the primary objective is often risk management, specifically, anticipating periods of poor air quality. The NNAR model's "flat" forecast underestimates future maximums, effectively hiding potential risk events. The ARIMA model, by maintaining the variance and amplitude of the series, provides a more realistic range of expected values. It allows the capability to anticipate not just the average air quality, but the worst-case scenarios likely to occur in specific months.

The stability of the ARIMA projection also suggests that the underlying data-generating process is linear and seasonal. The neural network, while capturing non-linearities in the training data, likely overfitted to short-term patterns and failed to generalize the seasonal recurrence over a multi-year horizon. While the NNAR model offers precision for immediate forecasting, the ARIMA model is the robust choice for the 72-month strategic horizon. Its ability to sustain the seasonal narrative of the data outweighs the marginal gain in short-term accuracy provided by

the neural network. For the long-term, the ARIMA forecast provides a safer and more physically interpretable baseline.

Chapter V – Discussion and Conclusion

A. Discussions

The results of this study highlight the strength of traditional statistical time series models, particularly SARIMA, in forecasting structured environmental indices such as the Baseline Impact-Weighted Philippine Air Quality Index. Among the candidate SARIMA models evaluated, SARIMA(6,1,1)(1,0,1)[12] emerged as the optimal specification based on AIC and comprehensive residual diagnostics, capturing both short-term autoregressive behavior and recurring seasonal cycles observed in the Philippine air quality series. This outcome closely aligns with findings from related literature, including Hasan et al. (2024), who also identified ARIMA-type models as the best-performing forecasting method for particulate matter concentrations in Dhaka. Their results emphasized that even in the presence of complex pollutant dynamics, linear time series models can outperform more flexible approaches when the underlying process exhibits strong periodicity and stable stochastic structure. The consistency between the present study and previous research underscores the suitability of SARIMA models for medium- to long-term air quality forecasting, where maintaining seasonal integrity and amplitude is crucial for realistic scenario generation and environmental risk interpretation.

B. Conclusions

A Baseline Impact-Weighted Philippine Air Quality Index was successfully formulated using population-weighted and covariance-adjusted long-term respiratory mortality relative risks.

Observed fluctuations also aligned with real-world atmospheric events, highlighting the index's responsiveness to actual conditions. Various forecasting models, including ARIMA, ETS, and neural networks, were trained and evaluated. The SARIMA(6,1,1)(1,0,1)[12], ETS(A,N,A), and NNAR(5,1,4) models demonstrated adequate performance, with residuals resembling white noise. Among them, the neural network model exhibited the highest short-term accuracy, while the ARIMA model performed best for longer-term projections. Overall, the BIWPhAQI forecasts demonstrate a generally stable level with a repeating seasonal pattern, suggesting a predictable stable air quality from a health impact perspective.

C. Recommendations

While the study successfully delivers a foundational analytical framework for the Philippines' first impact-weighted air quality index, its maximum utility depends on future research and implementation. The current index relies on generalized Relative Risks for mortality. To truly capture the complex health profile of the Philippine population, considering local demographics, unique pollutant mixtures, and healthcare capacity, it is strongly recommended to create dedicated, nation-specific epidemiological studies. This data is essential to derive localized Relative Risks that will ensure the BIWPhAQI is a valid and accurate reflection of local health threats.

The current modeling scope is also constrained by air quality data availability only up to the year 2021. For the BIWPhAQI to remain a relevant and predictive public health tool, its training and forecasting must integrate the most current data available. Future implementation orders the establishment of timely data acquisition pipelines to capture post-2021 trends, reflecting the impact of recent policy shifts and socio-economic changes.

Finally, it is also recommended that local environmental and public health agencies adopt the established methodology for the BIWPhAQI. This provides a clear, health-centric standard for monitoring. By treating this initial model as a dynamic foundation, it can be continuously refined and updated with both the localized Relative Risks and current air quality data, ensuring the Philippines has a powerful, evolving tool for health-protective air quality management.

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Appendix

Appendix A: Annotated R Code

```
#Load Packages
library(fpp2)
library(devtools)
library(GGally)
library(patchwork)
library(seasonal)
library(moments)
library(tseries)
library(lubridate)
library(dplyr)
library(tidyr)
library(fracdiff)
library(urca)

#Data Processing
df <- read.csv('climate_air_quality.csv')

#Converting each pollutant to the same metric
conversions <- c(no2 = 1.868,
                 o3  = 1.949,
                 so2 = 2.602,
                 co  = 1145)

df <- df %>%
  mutate(
    no2 = no2 * conversions["no2"],
    o3  = o3  * conversions["o3"],
    so2 = so2 * conversions["so2"],
    co  = co  * conversions["co"]
  )

#Taking the B coefficients from RR as weights
weights <- c(pm25 = log(1.14)/10,
             pm10 = log(1.12)/10,
             no2  = log(1.05)/10,
             o3   = log(1.05)/10,
             so2  = log(1.0067)/10,
             co   = log(1.0318)/1000)

#Computing the covariance-adjusted Index
pollutants <- df %>%
  select(pm25, pm10, no2, o3, so2, co) %>%
```

```

  drop_na()
Sigma <- cov(pollutants)
W <- diag(weights)
weighted_cov <- W %*% Sigma %*% W
X <- as.matrix(pollutants)
adj_index <- sqrt(rowSums((X %*% weighted_cov) * X))
df_clean <- df %>%
  drop_na(pm25, pm10, no2, o3, so2, co) %>%
  mutate(adj_index = adj_index)

df2 <- df_clean %>% select(adm4_pcode,date,adj_index)

#Importing population data
pop <- read.csv('worldpop_population.csv')

pop_ts <- pop %>%
  select(adm4_pcode, date, pop_count_total) %>%
  arrange(adm4_pcode, date) %>%
  mutate(date = as.Date(date))

# Forecast 2021 and 2022
pop_forecast <- pop_ts %>%
  group_by(adm4_pcode) %>%
  group_modify(~{
    df <- .x %>% filter(!is.na(pop_count_total))
    if(nrow(df) < 2) return(tibble(date = as.Date(character()), pop_count_total = numeric()))

    df <- df %>% mutate(year_num = year(date))
    fit <- lm(pop_count_total ~ year_num, data = df)
    new_dates <- tibble(year_num = 2021:2022)
    preds <- predict(fit, newdata = new_dates)

    tibble(date = as.Date(paste0(2021:2022, "-01-01")),
           pop_count_total = preds)
  }) %>%
  ungroup()

# Combine original and forecasted data
pop_combined <- bind_rows(pop_ts, pop_forecast) %>%
  arrange(adm4_pcode, date)

#Adding population data to air quality data
df2 <- df2 %>% mutate(date = as.Date(date))
pop_combined <- pop_combined %>% mutate(date = as.Date(date))
df2 <- df2 %>% mutate(year = year(date))
pop_combined <- pop_combined %>% mutate(year = year(date))

```

```

df3 <- df2 %>%
  left_join(pop_combined %>% select(adm4_pcode, year, pop_count_total),
    by = c("adm4_pcode", "year"))

#Weighted Averaging based on population
daily_index <- df3 %>%
  group_by(date) %>%
  summarise(
    adj_index_pop = sum(`adj_index` * pop_count_total, na.rm = TRUE) /
    sum(pop_count_total, na.rm = TRUE),
    .groups = "drop"
  )

#Monthly Averaging
df_monthly <- daily_index %>%
  mutate(month = floor_date(date, unit = "month")) %>%
  group_by(month) %>%
  summarise(
    adj_index_pop = mean(adj_index_pop, na.rm = TRUE),
    .groups = "drop"
  )
write.csv(df_monthly, "monthly_adj_index_fixed.csv", row.names = FALSE)

#Making a time series object
str(df_monthly[,2])
aqi_ts <- ts(unlist(df_monthly[,2]), start = c(2003, 1), freq = 12)

#Characterizing the AQI
autoplot(aqi_ts) + xlab("Year") + ylab("IB-Ph AQI") + theme_minimal()

summary(aqi_ts)

find_date <- function(value) {
  idx <- which.min(abs(aqi_ts - value))
  year <- 2003 + (idx - 1) %/% 12
  month <- 1 + (idx - 1) %% 12
  return(data.frame(value = aqi_ts[idx], idx = idx, year = year, month = month))
}

find_date(7.790); find_date(9.423); find_date(10.010);find_date(10.718);find_date(15.320)

skewness(aqi_ts); kurtosis(aqi_ts); sd(aqi_ts)

tsoutliers(aqi_ts)

plot(decompose(aqi_ts))

```

```

plot(decompose(aqi_ts,type='multiplicative'))
autoplot(stl(aqi_ts, s.window='periodic')) + ggtitle("STL decomposition") + theme_minimal()

ggseasonplot(aqi_ts) +
  ggtitle("Seasonal Plot of aqi_ts") +
  ylab("Temperature Anomaly") +
  xlab("Month") +
  theme(legend.position = "none") +
  theme_minimal()

ggAcf(aqi_ts) + theme_minimal()
ggPacf(aqi_ts) + theme_minimal()

summary(ur.kpss(aqi_ts))
lam <- BoxCox.lambda(aqi_ts)
aqi_transformed <- BoxCox(aqi_ts, lambda = BoxCox.lambda(aqi_ts))
summary(ur.kpss(diff(aqi_transformed)))
autoplot(diff(aqi_transformed)) + theme_minimal()

#Model Training
train<- window(aqi_ts,end=c(2020,12))
test <- window(aqi_ts,start=c(2021,1))

#ARIMA Selection
get.best.arima <- function(transformed_train, maxord=c(2,2,2,2,2,2))
{
  best.aic<-1e8
  n<-length(transformed_train)
  for (p in 0:maxord[1]) for (d in 0:maxord[2]) for (q in 0:maxord[3])
    for (P in 0:maxord[4]) for (D in 0:maxord[5]) for (Q in 0:maxord[6])
    {
      fit<-arima (transformed_train, order=c(p,d,q), seas=list(order=c(P,D,Q),
frequency(transformed_train)), method="CSS")
      fit.aic<--2*fit$loglik+(log(n)+1)*length(fit$coef)
      if (fit.aic<best.aic)
      {
        best.aic<-fit.aic
        best.fit<-fit
        best.model<-c(p,d,q,P,D,Q)
      }
    }
  }
  list(best.aic,best.fit,best.model)
}
get.best.arima(transformed_train)

```

```
(model_102_210 <- auto.arima(train,lambda = BoxCox.lambda(train),biasadj = T))
(model_200_101 <- Arima(train,order=c(2,0,0),seas=list(order=c(1,0,1),frequency(train)),
lambda = BoxCox.lambda(train),biasadj = T))
(model_111_100 <- Arima(train,order=c(1,1,1),seas=list(order=c(1,0,0),frequency(train)),
lambda = BoxCox.lambda(train),biasadj = T))
(model_611_100 <- Arima(train,order=c(6,1,1),seas=list(order=c(1,0,0),frequency(train)),
lambda = BoxCox.lambda(train),biasadj = T))
(model_611_101 <- Arima(train,order=c(6,1,1),seas=list(order=c(1,0,1),frequency(train)),
lambda = BoxCox.lambda(train),biasadj = T))
```

```
models <- list(
  m1 = model_102_210,
  m2 = model_200_101,
  m3 = model_111_100,
  m4 = model_611_100,
  m5 = model_611_101
)
```

```
sapply(models, AIC)
```

```
#NNAR Selection
```

```
transformed_train <- BoxCox(train, lambda = BoxCox.lambda(train))
```

```
AR1 <- Arima(
  transformed_train,
  order = c(1, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)
```

```
AR2 <- Arima(
  transformed_train,
  order = c(2, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)
```

```
AR3 <- Arima(
  transformed_train,
  order = c(3, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)
```

```
AR4 <- Arima(
  transformed_train,
  order = c(4, 0, 0),
```

```

seasonal = list(order = c(1, 0, 0), period = 12)
)

AR5 <- Arima(
  transformed_train,
  order = c(5, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)

AR6 <- Arima(
  transformed_train,
  order = c(6, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)

AR7 <- Arima(
  transformed_train,
  order = c(7, 0, 0),
  seasonal = list(order = c(1, 0, 0), period = 12)
)

AR_models <- list(
  AR1 = AR1,
  AR2 = AR2,
  AR3 = AR3,
  AR4 = AR4,
  AR5 = AR5,
  AR6 = AR6,
  AR7 = AR7
)

sapply(AR_models, AIC)

nnetar1 <- nnetar(transformed_train, p = 5, P = 1, size = 4)
nnetar1

#Fitting the best models for ARIMA, ETS, and NNAR
(fit1 <- Arima(train,order=c(6,1,1),seas=list(order=c(1,0,1),frequency(train))), lambda =
BoxCox.lambda(train),biasadj = T))
checkresiduals(fit1)
(fit2 <- ets(train,lambda = BoxCox.lambda(train),biasadj = T))
checkresiduals(fit2)
(fit3 <- nnetar(train,lambda = BoxCox.lambda(train),biasadj = T))

```

```

checkresiduals(fit3$residuals)

#Testing 2021-2022
fc1 <- forecast(fit1,h=24)
fc2 <- forecast(fit2,h=24)
fc3 <- forecast(fit3,h=24)

accuracy(fc1,test);accuracy(fc2,test);accuracy(fc3,test)

autoplot(train) + autolayer(fc1,series='arima',PI=FALSE) +
autolayer(fc2,series='ets',PI=FALSE) + autolayer(fc3,series='nnet') +
autolayer(test,series='test') + theme_minimal()
autoplot(test,series='test')+autolayer(fc1,series='arima',PI=FALSE) +
autolayer(fc2,series='ets',PI=FALSE) + autolayer(fc3,series='nnet') + theme_minimal()

#Forecasting 2021-2026
fc1.pt <- forecast(fit1,h=72)
fc2.pt <- forecast(fit2,h=72)
fc3.pt <- forecast(fit3,h=72)

accuracy(fc1.pt,test);accuracy(fc2.pt,test);accuracy(fc3.pt,test)

autoplot(train) + autolayer(fc1.pt,series='arima',PI=FALSE) +
autolayer(fc2.pt,series='ets',PI=FALSE) + autolayer(fc3.pt,series='nnet') +
autolayer(test,series='test') + theme_minimal()
autoplot(test,series='test')+autolayer(fc1.pt,series='arima',PI=FALSE) +
autolayer(fc2.pt,series='ets',PI=FALSE) + autolayer(fc3.pt,series='nnet') + theme_minimal()

```